

Course Outline



Basic Information

Term	Fall 2025
Course Title	Statistics and Probability
Course Code	STA 2101
Section	02
Credit Hours	3.0
Class Schedule	Sun/Tues: 04.30 PM - 05:50 PM
Pre-requisite Course	N/A
Program	B.Sc in CSE
Faculty	Atanu Shuvam Roy
Contact Email	atanu.shuvam@ulab.edu.bd
Contact Number	+8801937545261
Office and Location	Faculty Room-PC-325 (ULAB Permanent Campus)
Counseling/Office Hour	Sunday: 02:00-4:00 PM Note: Available by appointment (through email, text)
Google Classroom Link	
Number of Lectures	24

1.0 Course Description (from syllabus)/Rational:

This course contains statistics and probability which are fundamental knowledge for the Engineering students and for the students of Science and Technology. This course will introduce students to the fundamentals of the use of quantitative data. It will make students understand the facts expressed in quantitative form. It will also make the students able to apply statistical techniques in solving Engineering problems.

1.1 Course Objectives:

- To provide a thorough understanding the meaning of Statistics and Probability with real-life applications and the resource requirements.
- To introduce the various methods of statistical data in an investigation.
- To enable students to apply the techniques of statistical and probability methods to identify the problems.
- To emphasize on efficient for solving practical problems through various formulae to be used in the implementations of technical problems.
- To expose the students to a variety of techniques that have practical applications, while conducting detailed analysis of the requirements.

1.2 Mapping of Course Outcomes (COs) to Program Outcomes (POs)

CO	Description	Corresponding PO No.	K/P/A	Bloom's Taxonomy Domain/Level	Delivery Methods and Activities	Assessment Tools
CO1	Understand the fundamentals of Statistics & Probability	-	-	Cognitive/L2, Affective/L3	Lecture, Class Participation	Mid-term exam, Final exam, Assignment/Project
CO2	Analyze a dataset including both categorical and quantitative variables to support or refute a statement	1	K2, K4/ P1, P3	Cognitive/L3, Affective/L3	Lecture, Class Participation	Mid-term exam, Final exam, Assignment/Project
CO3	Perform testing and statistical inference in several circumstances and interpret the results in an applied context.	4	K2, K4, K8/ P1, P3, P6	Cognitive/L5	Lecture, Class Participation	Assignment/Project

1.3 Teaching and Learning Activities (TLA)

TLA1	Interactive discussion using Online/multimedia or whiteboard.
TLA2	Interactive video and/or scenario based presentation
TLA3	Case Study and group discussion
TLA4	Real-life project in a team/single to apply design knowledge

1.4 Course Delivery Plan (include Lab if any)

Week	Discussion Topic and Book	Delivery methods and Activities	Assessment and Mapping with CO
Wk 1	Lesson-1: Understanding Statistics, Descriptive and Inferential Statistics, Achieving the objective of Inferential Statistics, Variable, Experimental Unit, Measurement or Data, Population, Sample, Univariate, Bivariate and Multivariate Data Lesson-2: Types of Variables, Qualitative, Quantitative variable, Discrete and Continuous Variable, Graphs for Categorical Data, frequency, relative frequency, percentage, pie chart and bar charts for data distribution	Lesson-1 & 2: Online/Onsite discussion; Using PPT; TLA1	Background Preparation week
Wk 2	Lesson-3: Graphs for Quantitative Data, Categorical and Numerical data - pie chart and bar chart, time series data- line chart, dot plot,	Lesson-3 & 4: Online/Onsite discussion; Using PPT; TLA1, TLA2 Tools to use: jupyter Notebook	CLO1

	stem and leaf plot, Relative Frequency Histograms Lesson-4: Describing a set of data with numerical measures, Measures of Center : Arithmetic Mean, Median, Mode	Hands-on: Working with Google Colab for basic python; Working with Numpy and Panda	
Wk 3	Lesson-5: Measures of Variability, range, variance of population, variance of sample, standard deviation, practical significance of standard distribution Lesson-6: Measures of Relative Standing: Z score, percentile, quartile, interquartile range, The five number summary and box plot	Lesson-5 & 6: Online/Onsite discussion; Using PPT; TLA1, TLA2, TLA4 Hands-on: Working for data pre-processing using Google colab; Student form team for the project and fill the team info using Google form.	CLO1, CLO3
Wk 4	Lesson-7: Describing Bivariate Data, Graphs for Categorical variables, Side by side pie charts, stacked bar charts, conditional data distributions, Scatterplots for two quantitative variables Lesson-8: Numerical measurements for Quantitative bivariate data	Lesson-7 & 8: Online/Onsite discussion; Using PPT; TLA1, TLA2. Tools to use: jupyter Notebook Hands-on: Working with Numpy and Pandas, Matplotlib	CLO1 Class Test# 1
Wk 5	Lesson-9: Correlation coefficient, covariance, linear relationship, regression or least squares line Lesson-10: working with python notebook	Lesson-9 & 10: Online/Onsite discussion; Using PPT; TLA1, TLA2. Hands-on: Working with Python packages	CO1, CLO3
Wk 6	Lesson 11: working with python notebook Lesson 12: working with python and numpy and pandas	Lesson-11 & 12: Online/Onsite discussion; Using PPT; TLA1, TLA2. Hands-on: Working with Python packages	CO1
Wk 7	Midterm Exam Week		
Wk 8	Lesson 13: working with python and numpy and pandas Lesson 14: Probability and probability distributions, Events and sample space, tree diagram, probability table, ven diagram, calculating probabilities of event (MID Project Feedback session)	Lesson-13 & 14: Online/Onsite discussion; Using PPT; class work, TLA1, TLA2, TLA3, TLA4.	CO2

Wk 9	Lesson 15: Event relations and probability rules, Independence, multiplication rule, conditional probability Lesson 16: Bayes Rule and total probability	Lesson-15 & 16: Online/Onsite discussion; Using PPT; class work, TLA1, TLA2, TLA3, TLA4. Hands-on: Working with Python packages	CO2 , CO3
Wk 10 2nd CT Week	Lesson 17: Discrete random variable and their probability distributions, mean and standard deviation Lesson 18: Binomial probability distribution, mean and standard deviation for binomial random variable	Lesson-17 & 18: Online/Onsite discussion; Using PPT; class work, TLA1, TLA2, TLA3, TLA4. Hands-on: Working with Python packages	CO2 Class Test# 2
Wk 11	Lesson 19: Poisson Probability Distribution Lesson 20: Normal Probability Distribution for continuous random variable	Lesson-19 & 20: Online/Onsite discussion; Using PPT; class work, TLA1, TLA2, TLA3, TLA4. Hands-on: Working with Python packages	CO2
Wk 12	Lesson 21: Tabulated Areas of Normal distribution Lesson 22: Sampling Plans and Experimental Designs, Statics and Sampling Distributions, The central limit theorem, Sampling distribution of the sample mean, sample proportion, A Sampling Application: Statistical Process Control	Lesson-21 & 22: Online/Onsite discussion; Using PPT; class work, TLA1, TLA2, TLA3, TLA4.	CO3
Wk 13	Lesson 23: Final Project presentation Lesson 24: Final Project presentation and Discussion.	Lesson-23 & 24: TLA2 and TLA4. Hands-on: Project presentation.	CO3 Project report submission
Wk 14 and Wk 15	Final Exam Week		

1.5 Text and Reference Materials

- **Text Book(s):** Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver

Reference Material/Book(s):

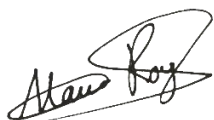
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross

1.6 Distribution of Marks for Assessment

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

1.7. Grading Policy

Policy	Letter Grade	Grade Point	Assessments
95% and above	A+	4.00	Outstanding
85% to 94%	A	4.00	Superlative
80% to 84%	A-	3.80	Excellent
75% to 79%	B+	3.30	Very Good
70% to 74%	B	3.00	Good
65% to 69%	B-	2.80	Average
60% to 64%	C+	2.50	Below Average
55% to 59%	C	2.20	Passing
50% to 54%	D	1.50	Probationary
below 50%	F	0.00	Fail
--	I	0.00	Incomplete
--	W	0.00	Withdrawn
--	AW	0.00	Administrative Withdrawal



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 Course Coordinator/ Teacher
 Date: 28/05/2024

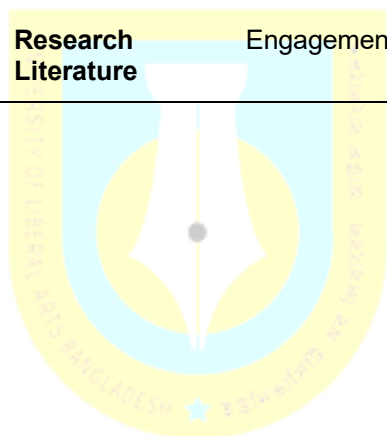
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 Head of the Department
 Date: 28/05/2024

Appendix-1: Program outcomes

POs	Program Outcomes
PO1	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.
PO3	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.
PO4	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
PO5	Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
PO8	Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.
PO9	Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.
PO10	Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.
PO12	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Knowledge Profile

K	Attribute	Description
K1	Natural Sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2	Mathematics	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modelling applicable to the discipline
K3	Engineering Fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
K4	Specialist Knowledge	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
K5	Engineering Design	Knowledge that supports engineering design in a practice area
K6	Engineering Practice	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
K7	Comprehension	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability
K8	Research Literature	Engagement with selected knowledge in the research literature of the discipline



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